

State Machine Question (100 marks)

Given the problem description below, specify the *state machine for a train* in the Advanced Train Automation System (ATAS). **Show the states as well as transitions including events, guards, and actions. Define composite states and orthogonal regions when needed to reduce complexity.** You do **not** need to specify parameters for events, guards, and actions. Use **meaningful** names for the private methods for guards and actions in your state machine. You do **not** need to define the method bodies of those methods.

Problem Description: The Advanced Train Automation System (ATAS) handles the movement of trains across the rail network, i.e., the trains are driverless. The control of the traffic signals, the opening and closing of gates at rail crossings, the setting of turnouts (railroad switches), and the dynamic adjustment of the train's itinerary are out of scope as they are handled by different systems. A train is scheduled several weeks before its departure date, so that passengers can book tickets for the train. The booking process is also out of scope.

The rail network is split into non-overlapping segments. For each segment, the desired train speed is defined. Three types of segments exist: a regular segment, a traffic light segment, and a station segment. The traffic light and station segments also define the coordinates of the location where the train is required to stop (if the light is red or if the station is on the train's itinerary, respectively). While the segments are of variable length, a traffic light or station is always at the end of its segment to allow a train to stop in time if required (i.e., the decision to stop is made when entering the segment). Each traffic light segment or station segment is always followed by a regular segment.

Each train is equipped with three systems that operate continuously and independently of the train as long as the train is active: a Segment Detection System (SDS), a Movement Detection System (MDS), and an Obstruction Detection System (ODS). Each of these systems runs checks every second when a train is active. The SDS detects if the train has entered a new segment and sets the current segment of the train accordingly. The MDS detects if the train is moving or not. The ODS detects whether the railway tracks are obstructed or not and initiates an emergency stop if they are.

At the beginning of its scheduled stop duration, ATAS powers on a train that is waiting at its first station so that the current segment is set to the first segment of the train's itinerary. After the stop duration, ATAS sets the required speed of the train. When the train enters a new segment, the required speed for the segment is set. However, if the segment is a traffic light segment and the light is red, ATAS instead initiates the stopping of the train at the traffic light location. Similarly, if the segment is a station segment, ATAS initiates the stopping of the train at the station location. The actual stopping of the train is confirmed by the MDS in both cases. At a red traffic light, the train waits until the light turns green at which point the ATAS sets the required speed of the train. At a station, the ATAS opens the train doors. The train then waits the scheduled stop duration after which the ATAS closes the train doors and sets the required speed. When the train arrives at its final station, the ATAS waits the scheduled stop duration before powering off the train.

In the case of an emergency, the train stops immediately. You may assume that the train stops in the same segment where an obstruction was detected. Once the train is cleared, the train continues its journey where it left off.

The train is controlled with the help of the following API: **setCurrentSegment(segment)**, **setSpeed(currentSegment.speed)**, **stopAtLocation(currentSegment.location)**, **emergencyStop()**, **powerOn()**, and **powerOff()**. You may use the following abbreviations in your solution: **setCS**, **setSpeed**, **stopAL**, **eStop**, **pOn**, and **pOff**, respectively.